

Final Project Deck

Leveraging Visibility Graphs in ML trading strategies

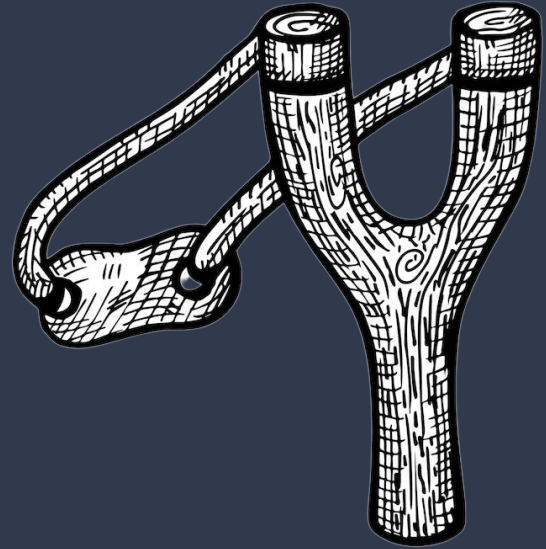
Motivation



Motivation

- A small, consistent edge moves enormous value in markets
- Yet most active strategies fail to beat passive benchmarks
- We wanted to test this ourselves — not just read about it
- Real market data forces us to confront ML pitfalls head-on

How Can We Leverage ML to Beat the Market?



Active Trading Strategies

Mean Reversion

Statistical Arbitrage

Swing Trading

VG Leveraged Trading

The Question



The Million Dollar Question

- Can ML find a predictable edge from price and volume alone?
- Most approaches use the same toolbox — lags, moving averages, volatility
- Our angle: read the **shape** of a price series, not just its numbers
- Visibility graph → ASPL as a structural feature
- Three tests: ASPL only / traditional only / combined — all vs. Buy & Hold

Related Work



Related Work

Study	Method	Result	Return backtest?
Li et al. 2023	LSTM + MACD/KDJ	+17.9% vs +9.7%	✓
Nordfjell & Ring 2023	Random Forest	+12.6% vs -8.8% CAGR	✓
Bielinskyi et al.	NVG + RF	58.4% vs 50% accuracy	✗
Xiu et al. 2021	VG crash detection	Beat VG-only baseline	✗
Wen et al. 2022 (review)	VG literature survey	—	✗

ML beats buy-and-hold. VG predicts well. **No one has tested VG on actual returns.**

The Data

Data

UC Berkeley Library

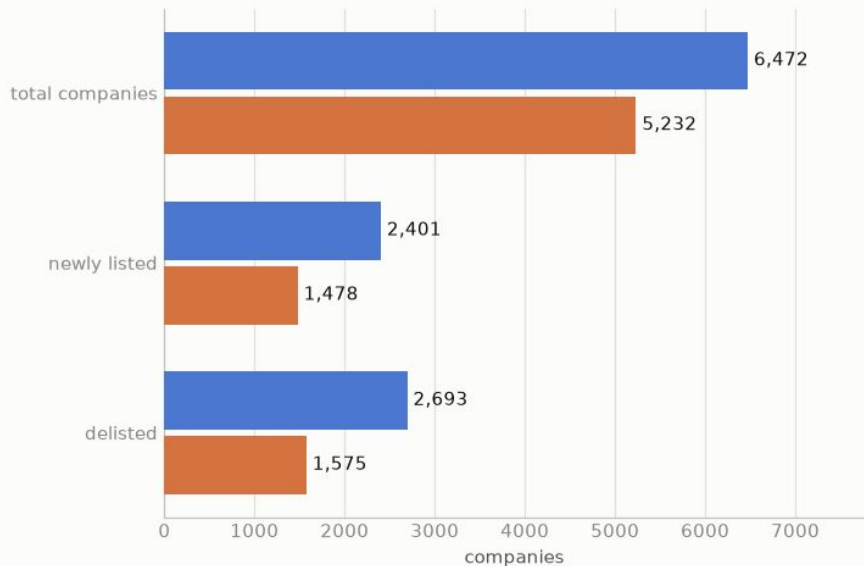


UC Berkeley
School of Information

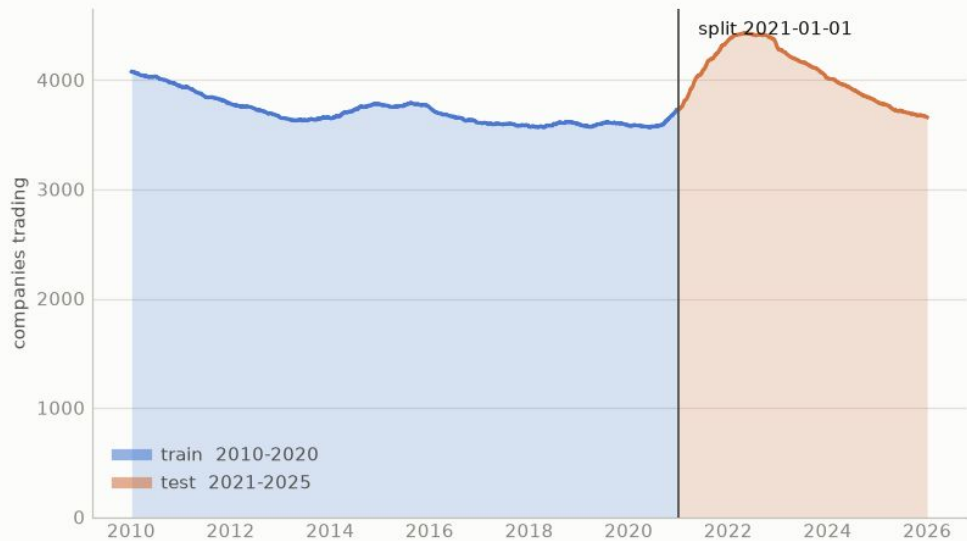


Data

The universe, and its churn

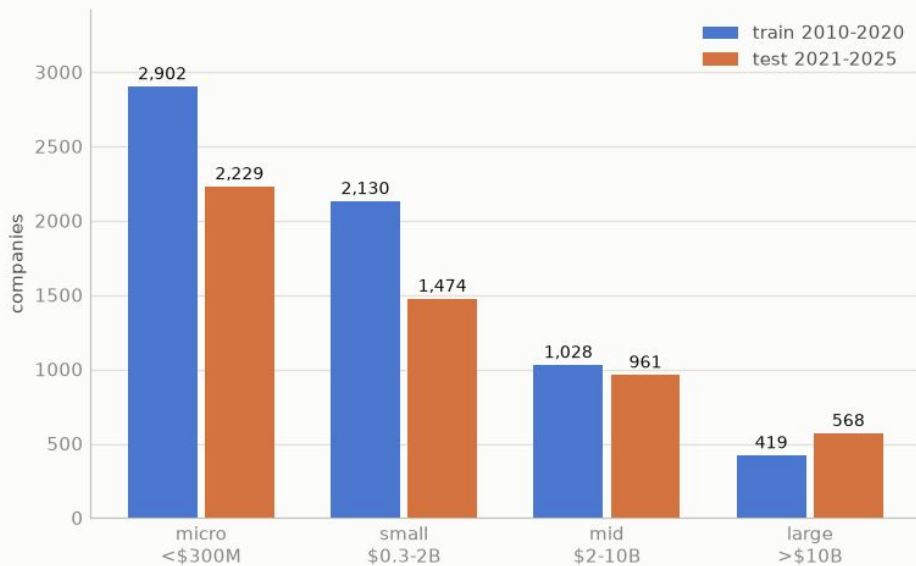


The universe over time, and where we split it

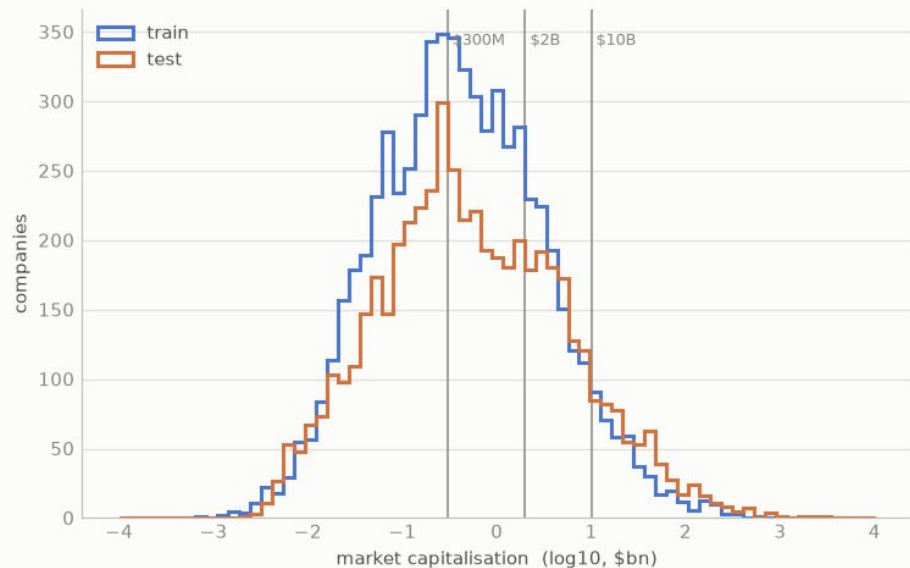


Data

The data is mostly very small companies

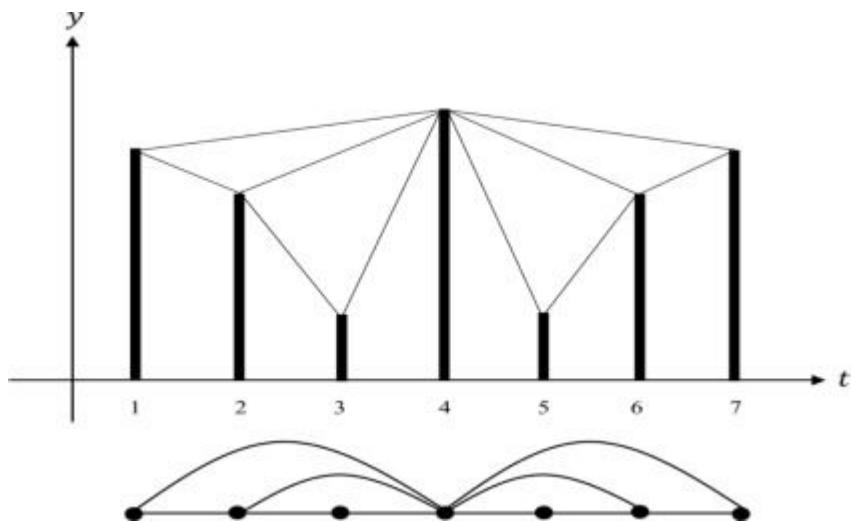


Size distribution, train vs test



Primer: Visibility Graphs

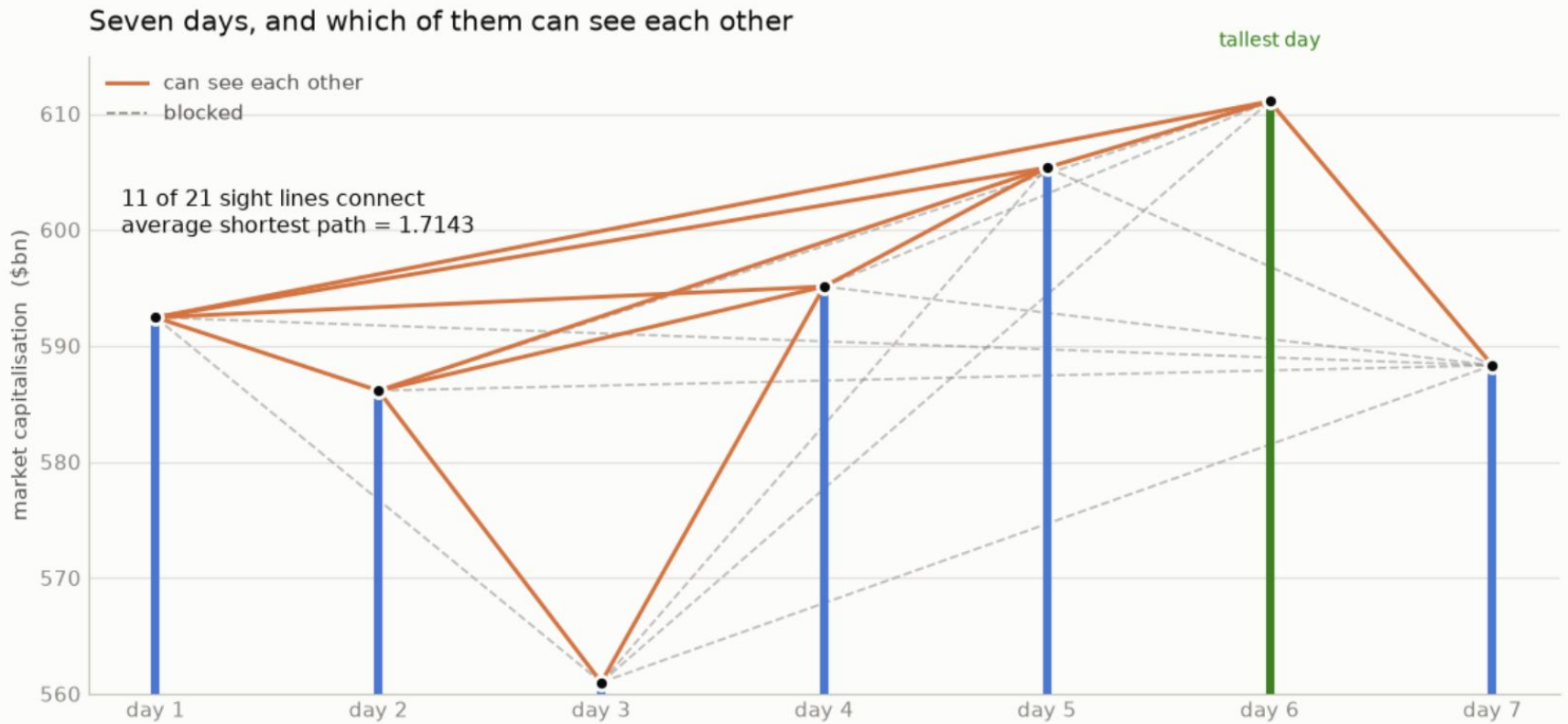
Numbers to Nodes



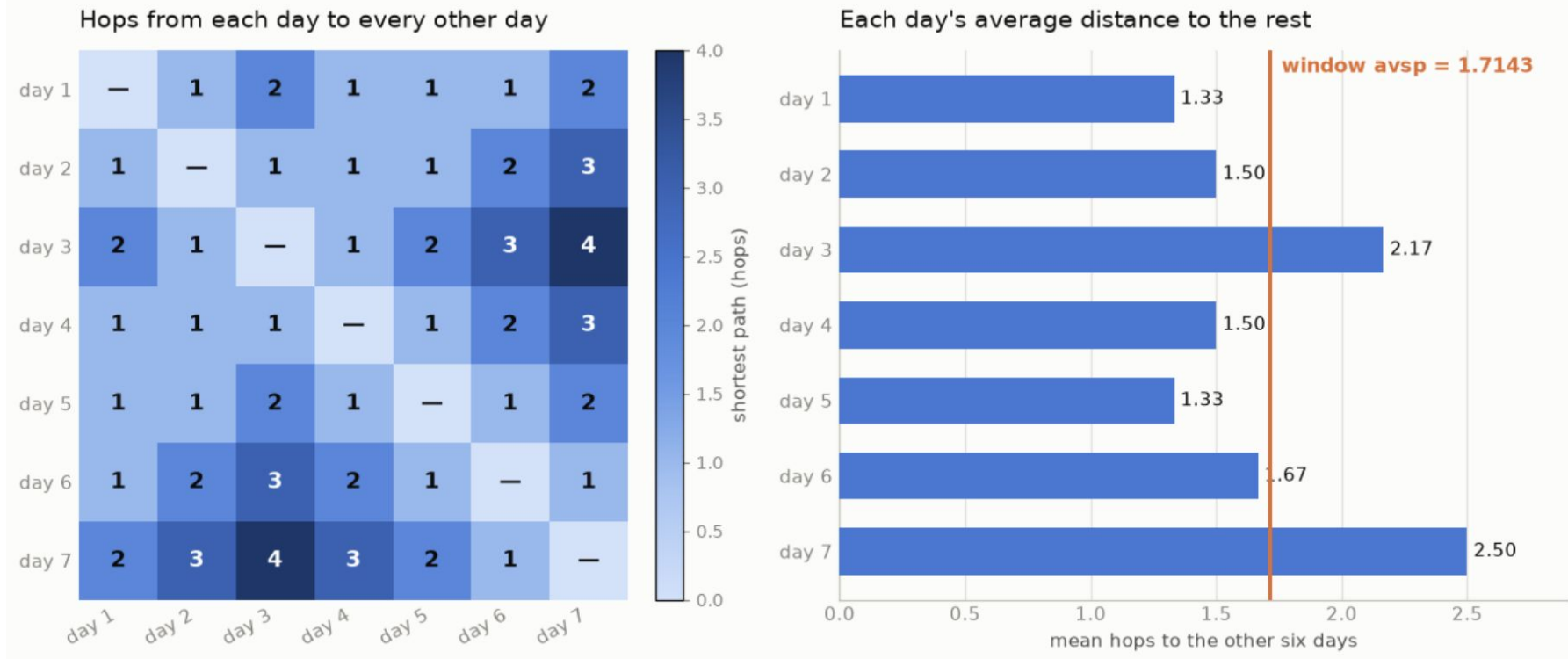
Big Idea

- Treats price bar data as connected nodes using the “Line of Sight” Rule
- Graphical structure unlocks graphical algorithm-created features
- The feature we leverage is Average Shortest Paths (**AVSP**) based on a specific rolling window (*more on window optimization later*)

Numbers to Nodes



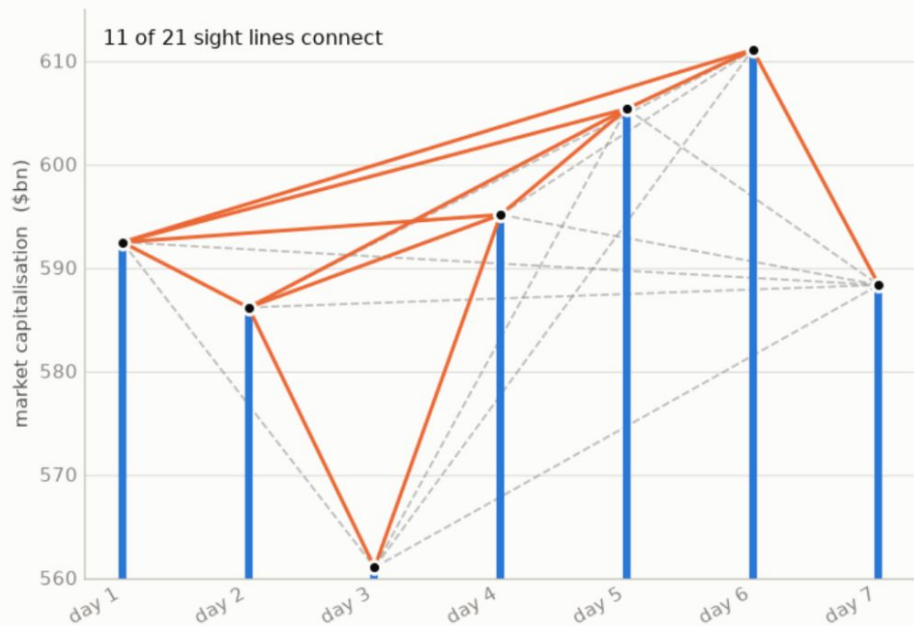
Numbers to Nodes



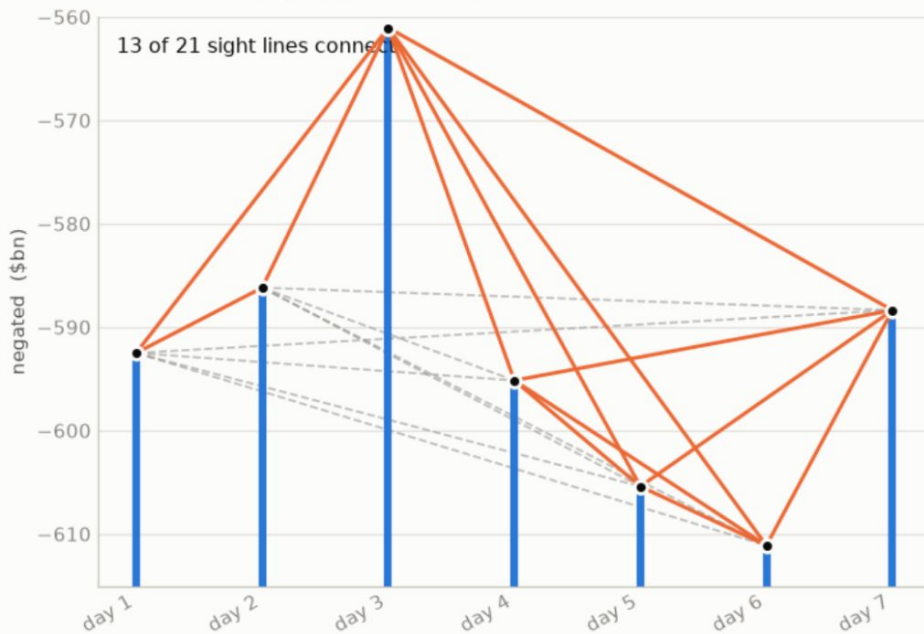
Numbers to Nodes

The same seven days, measured twice

as they happened pos = 1.7143



the same days, flipped neg = 1.3810



Methods

Trading Strategy Pipeline



Standardized Stock Price Data



Signal Generated



"All In or All Out"
(Full Entry or Exit Executed)

Different Signal Formation Approaches

Baseline: Random Signal

Pure VG Signal

VG X ML Signal

ML Regressor

ML Classifier

Trading Strategy Guidelines

- Initial Capital: We start with \$1000 for each stock.
- Full Entry/Exit:
 - Buy Signal: Use **all available capital** for a stock to buy shares.
 - Sell Signal: Sell **all shares** of a stock.

Baseline: Random Signal

Baseline

Pure VG

VG X ML



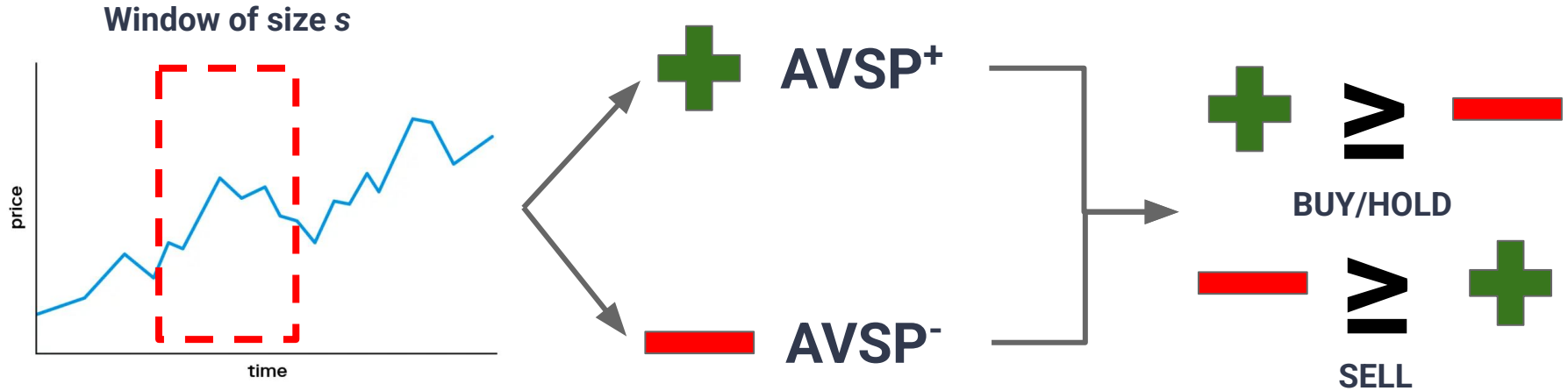
1. Choose a random number from a normal distribution.
2. If the selected number is greater than zero → **Buy Signal**
3. Otherwise **Sell Signal**

Pure VG Strategy

Baseline

Pure VG

VG X ML

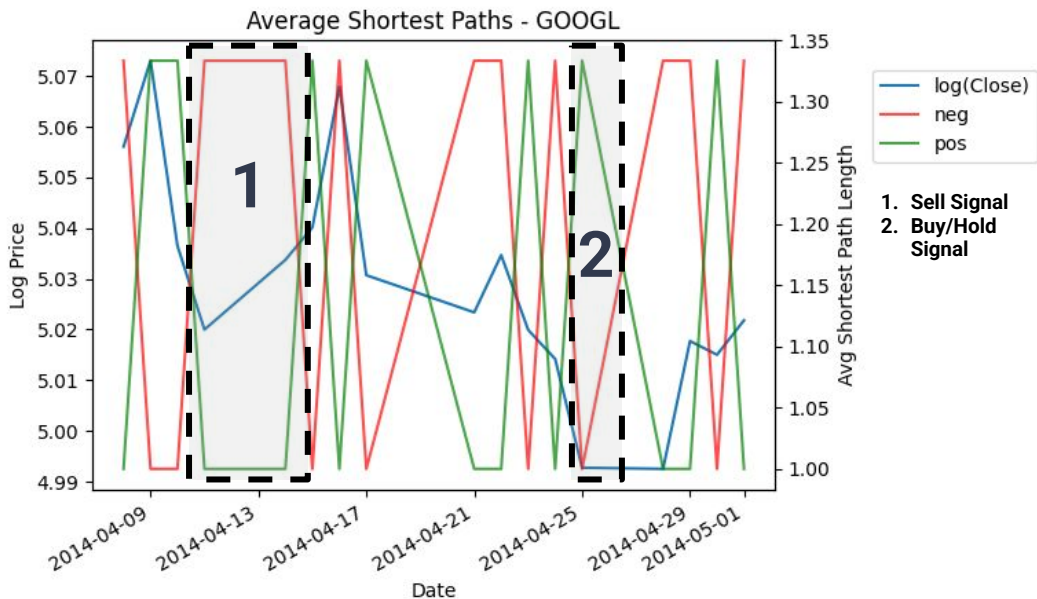


Pure VG Strategy - *Example*

Baseline

Pure VG

VG X ML

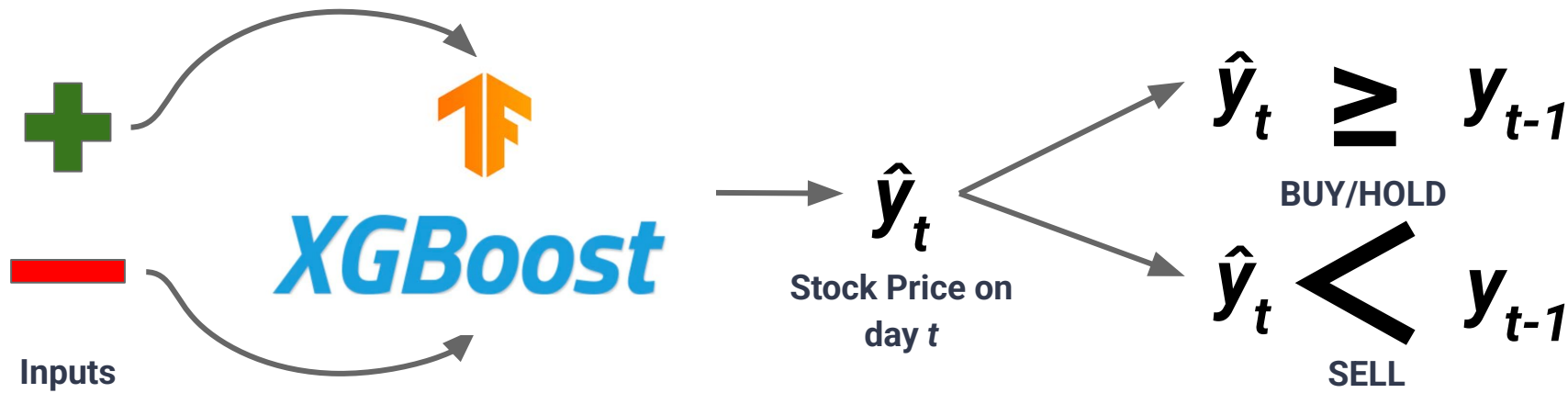


Models 1 & 2 – ML Regressors

Baseline

Pure VG

VG X ML



Models 3 & 4 – ML Classifiers

Baseline

Pure VG

VG X ML



$$\begin{array}{l} y_t \geq y_{t-1} \\ \sigma_t = 1 \\ y_t < y_{t-1} \\ \sigma_t = 0 \end{array}$$

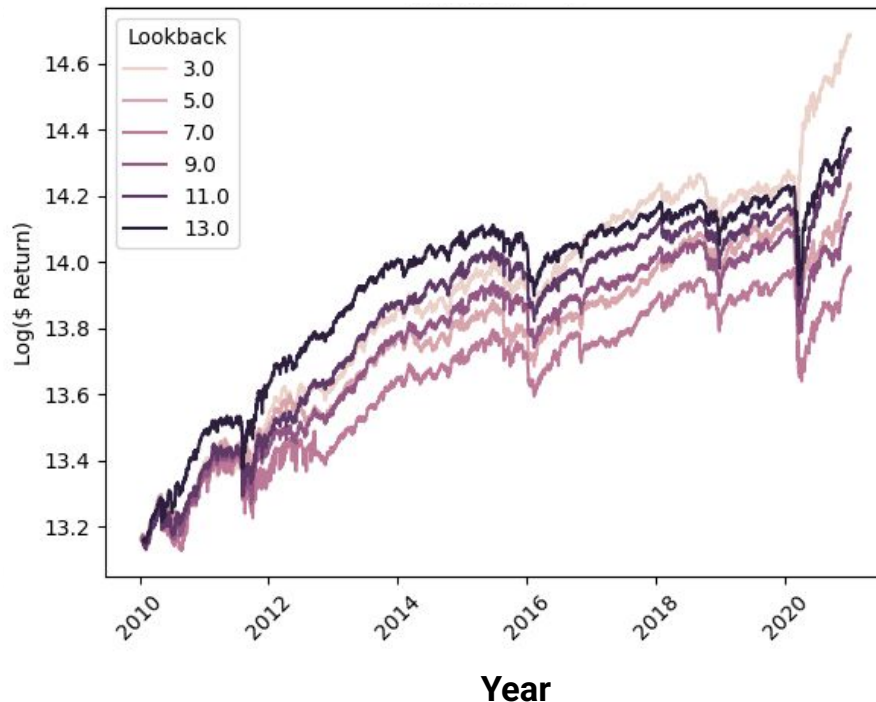
Experiments



Window Optimization

* After testing various lookbacks, a lookback window of **3** rendered the highest returns on the training data.

Portfolio Returns w/ Different Lookbacks



Model Experimentation/Fine-Tuning

Tensorflow Models

Tuned Parameters

- # Layers
- Layer Size
(# nodes)
- Dropout
- Batch Size

XGBoost Models

Tuned Parameters

- # Estimators
- Max Depth
- Learning Rate
- Subsample
- Colsample by Tree
- Min Child Weight

Objective Functions

Regressor: MSE

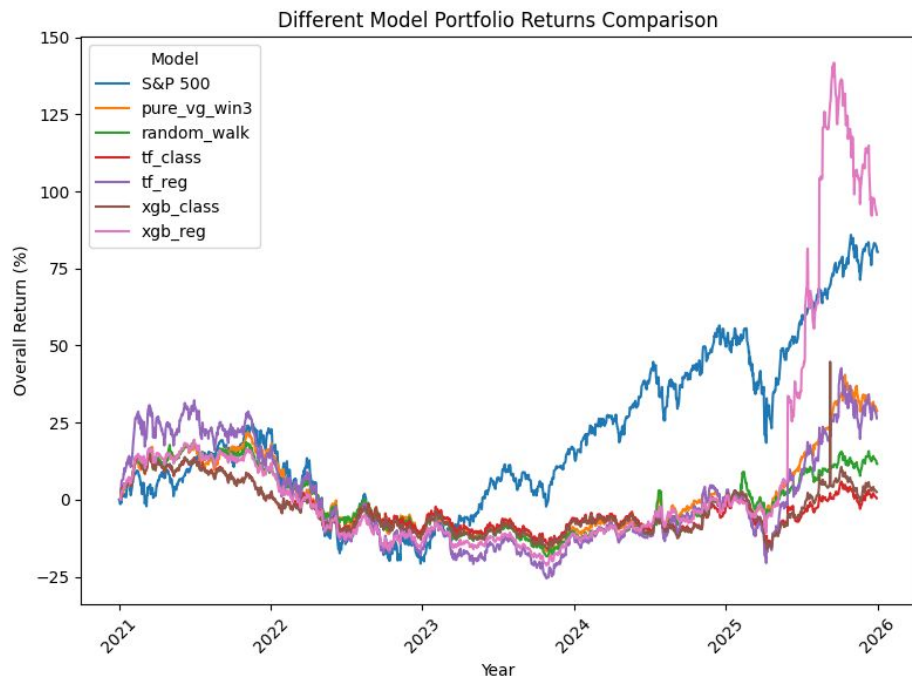
Classifier: AUC



Results



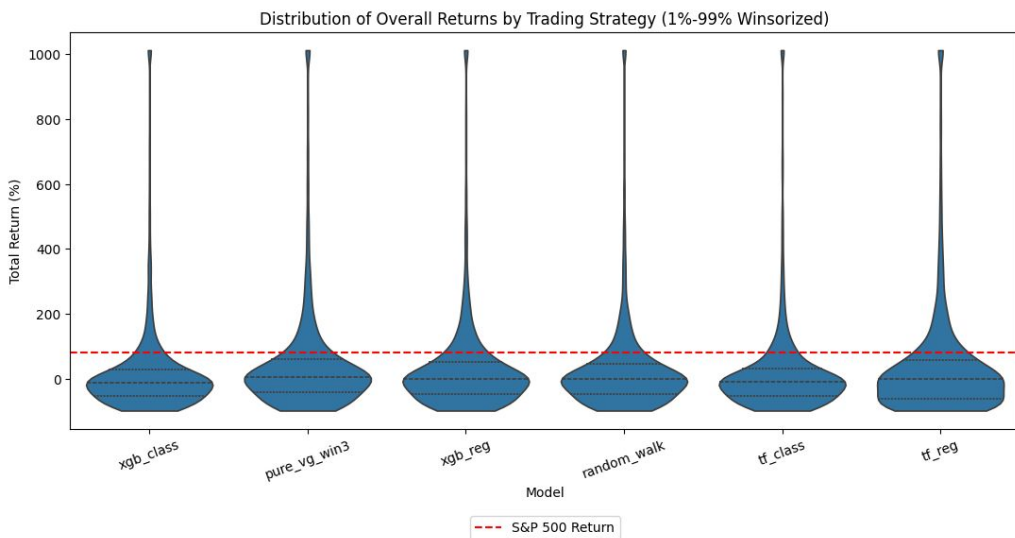
Different Model Returns Comparison



Model	Total Portfolio Growth
S&P 500	80%
pure_vg_win3	29%
random_walk	12%
tf_class	0%
tf_reg	26%
xgb_class	2%
xgb_reg	92%

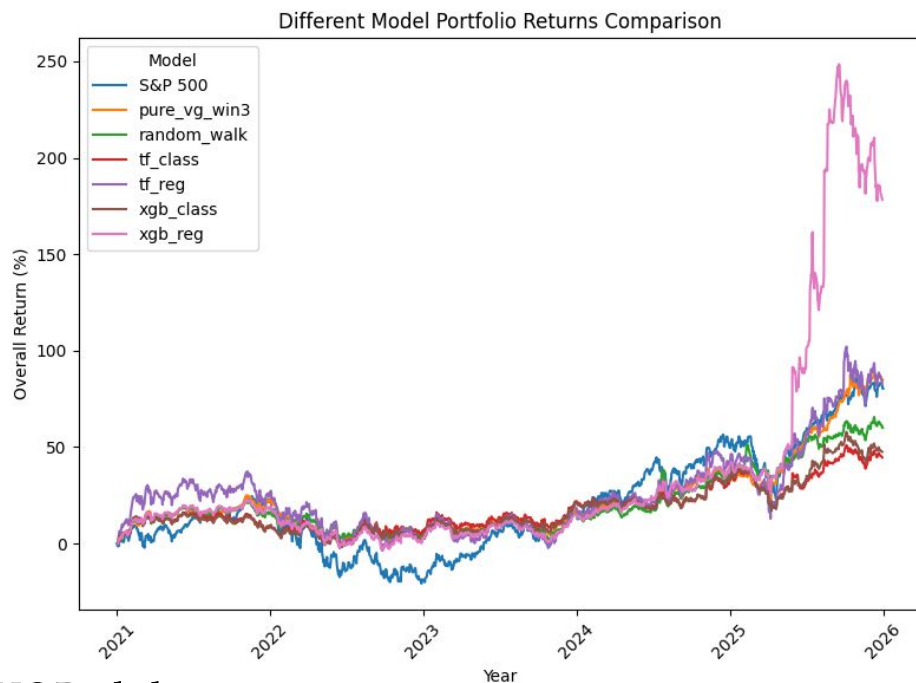
* The XGBoost Regressor Signal had the largest portfolio return, outperforming the S&P 500 and other models.

Individual Stock Return Dispersions



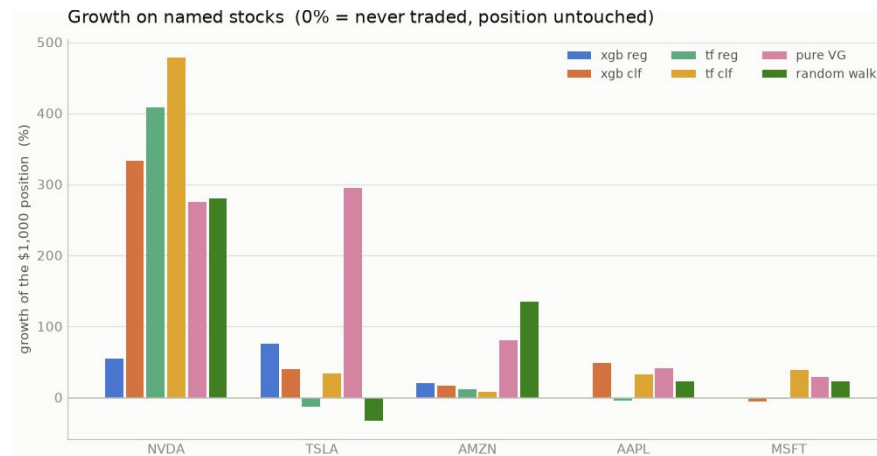
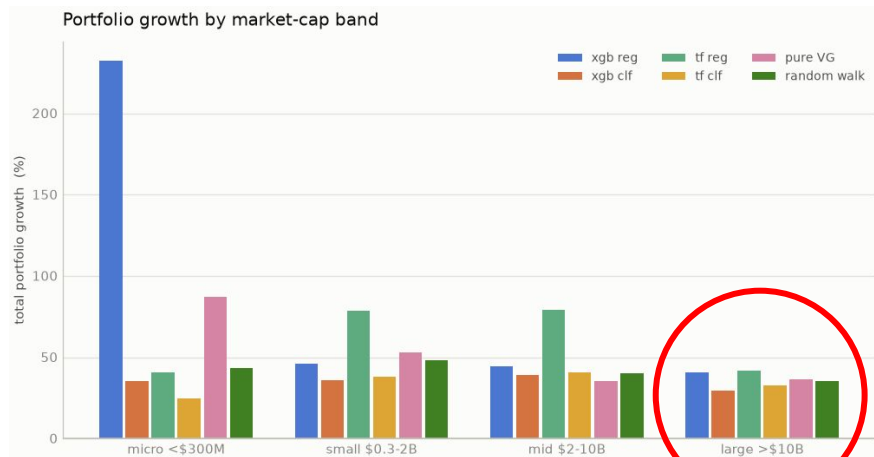
Model	Median Stock Return [Q1, Q3]
S&P 500	80%
pure_vg_win3	5% [-41%, 59%]
random_walk	-2% [-46%, 45%]
tf_class	-11% [-52%, 30%]
tf_reg	-2% [-62%, 58%]
xgb_class	-12% [-52%, 28%]
xgb_reg	-2% [-48%, 50%]

Subgroup Analysis – Survivor Stocks



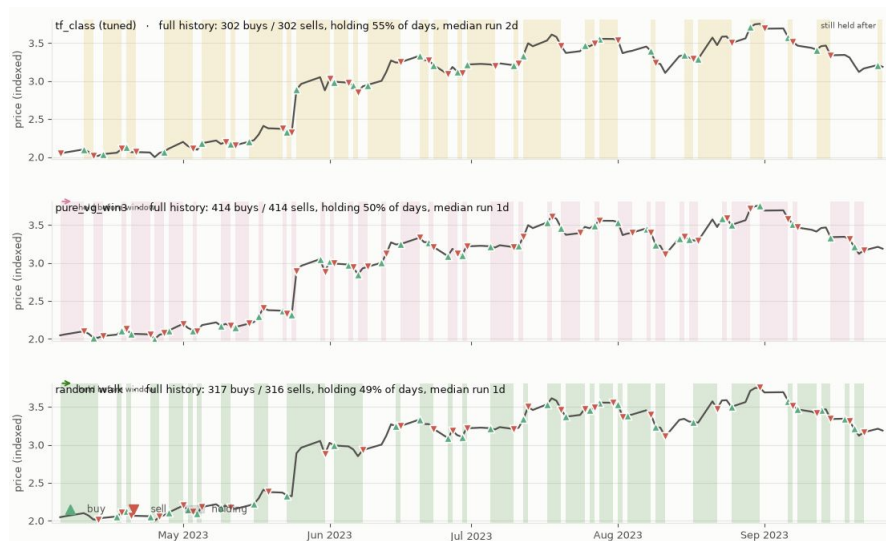
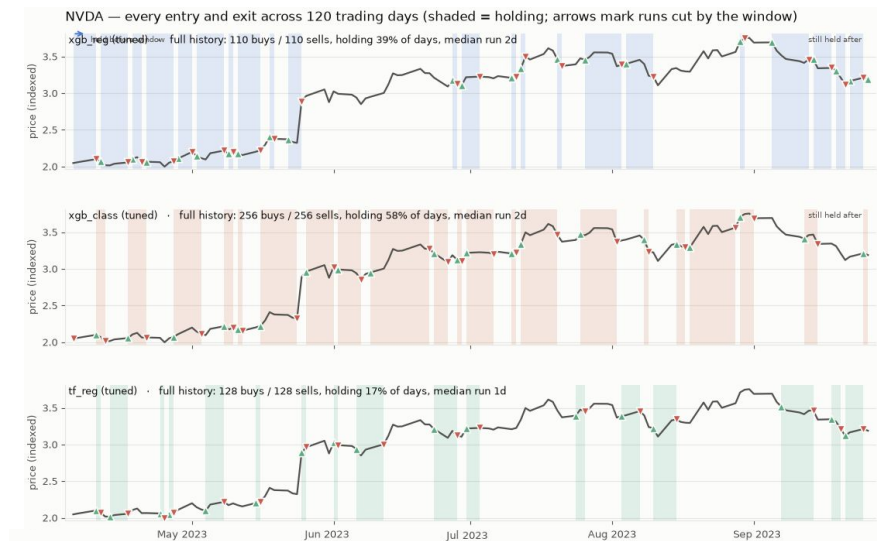
Model	Total Portfolio Growth (<i>All Stocks</i>)	Total Portfolio Growth (<i>Survivor Stocks</i>)	Survivor Stocks v. All Stocks Var
S&P 500	80%	80%	-
pure_vg_win3	29%	85%	+56%
random_walk	12%	60%	+48%
tf_class	0%	44%	+44%
tf_reg	26%	83%	+57%
xgb_class	2%	48%	+46%
xgb_reg	92%	178%	+86%

Different Cap Size, Different Growth



Zooming in on large
cap stocks...

Cap Size Matters – NVIDIA Case Study



Model Generalizability

Accuracy

Model	Train Set	Test Set	Test v. Train Var
pure_vg_win3	51%	51%	0%
tf_class	49%	49%	0%
tf_reg	51%	50%	-1%
xgb_class	49%	49%	0%
xgb_reg	50%	50%	0%

Precision

Model	Train Set	Test Set	Test v. Train Var
pure_vg_win3	53%	51%	-2%
tf_class	51%	49%	-2%
tf_reg	52%	50%	-2%
xgb_class	52%	49%	-3%
xgb_reg	52%	50%	-2%

Recall

Model	Train Set	Test Set	Test v. Train Var
pure_vg_win3	52%	51%	-1%
tf_class	48%	47%	-1%
tf_reg	95%	93%	-2%
xgb_class	52%	47%	-5%
xgb_reg	53%	55%	+2%

Model Selection

XGBoost

Regressor

Conclusion



Final Words

Active Trading Using a VG X ML
Model CAN beat the market

Practical Limitations

Equal investment ALL stocks

Unequal returns across different market cap size bands

Next Steps

Use VG X ML models to inform portfolio allocation, not just
entry/exit strategy

Develop a trading strategy that performs well across all market cap
size bands

THANK YOU

